

HDOC Day-40

Menu-driven C program using separate user-defined functions to perform string operations on a student's first, middle, and last name. Call by reference is used wherever a function modifies a string or returns results through parameters.

1. Problem Understanding

The program reads the student's first name, middle name, and last name. It repeatedly displays a menu and performs the selected string operation using separate functions. The program uses string.h functions such as strcpy(), strcat(), strlen(), strcmp(), and strstr(), and ctype.h functions such as isalpha(), isdigit(), isspace(), and tolower(). Functions that return multiple results use pointer parameters.

2. Standard Test Cases

S.No.	Complete Input	Expected Output
1	First: Aarav; Middle: Kumar; Last: Sharma; Choice: 1	Full name: Aarav Kumar Sharma
2	First: Aarav; Middle: Kumar; Last: Sharma; Choice: 2	Length of full name: 18
3	First: Aarav; Middle: Kumar; Last: Sharma; Choice: 3	Copied name: Aarav Kumar Sharma
4	First: Aman; Middle: Raj; Last: Aman; Choice: 4	First name and last name are equal
5	First: Aarav; Middle: Kumar; Last: Sharma; Choice: 5	Reverse name: Sharma Kumar Aarav
6	First: Aarav; Middle: Kumar; Last: Sharma; Choice: 6	Vowels: 7; Consonants: 9; Digits: 0; Spaces: 2
7	First: Aarav; Middle: Kumar; Last: Sharma; Choice: 7; Word: Kumar	Word found in full name
8	First: Aarav; Middle: Kumar; Last: Sharma; Choice: 7; Word: Singh	Word not found in full name
9	First: Aarav; Middle: Kumar; Last: Sharma; Choice: 8	Program exits

3. Algorithm

1. Start the program and read first name, middle name, and last name.
2. Use combineNames() to create the full name using strcpy() and strcat().
3. Display the menu repeatedly until the user selects Exit.
4. For choice 1, display the full name.
5. For choice 2, call findLength() and return the length through a pointer.
6. For choice 3, call copyName() to copy the full name into another string.

7. For choice 4, call compareNames() and return the strcmp() result through a pointer.
8. For choice 5, call reverseNameOrder() to create last name, middle name, first name order.
9. For choice 6, call countCharacters() and return vowels, consonants, digits, and spaces through pointers.
10. For choice 7, read a word, call searchWord(), and return whether strstr() finds it.
11. For choice 8, stop the program. For any other choice, display an invalid-choice message.

4. C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void combineNames(const char firstName[], const char middleName[], const char lastName[], char
fullName[])
{
    strcpy(fullName, firstName);
    strcat(fullName, " ");
    strcat(fullName, middleName);
    strcat(fullName, " ");
    strcat(fullName, lastName);
}

void findLength(const char fullName[], size_t *length)
{
    *length = strlen(fullName);
}

void copyName(const char source[], char destination[])
{
    strcpy(destination, source);
}

void compareNames(const char firstName[], const char lastName[], int *comparisonResult)
{
    *comparisonResult = strcmp(firstName, lastName);
}

void reverseNameOrder(const char firstName[], const char middleName[], const char lastName[], char
reverseName[])
{
    strcpy(reverseName, lastName);
    strcat(reverseName, " ");
    strcat(reverseName, middleName);
```

```

    strcat(reverseName, " ");
    strcat(reverseName, firstName);
}

void countCharacters(const char fullName[], int *vowels, int *consonants, int *digits, int *spaces)
{
    int i;
    char ch;
    *vowels = 0;
    *consonants = 0;
    *digits = 0;
    *spaces = 0;
    for (i = 0; i < strlen(fullName); i++)
    {
        ch = tolower(fullName[i]);
        if (isalpha(ch))
        {
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
                (*vowels)++;
            else
                (*consonants)++;
        }
        else if (isdigit(ch))
            (*digits)++;
        else if (isspace(ch))
            (*spaces)++;
    }
}

void searchWord(const char fullName[], const char word[], int *searchResult)
{
    if (strstr(fullName, word) != NULL)
        *searchResult = 1;
    else
        *searchResult = 0;
}

int main()

```

```

{
    char firstName[50], middleName[50], lastName[50];
    char fullName[150], copiedName[150], reverseName[150], word[50];
    int choice, comparisonResult, vowels, consonants, digits, spaces, searchResult;
    size_t length;
    printf("Enter first name: ");
    scanf("%49s", firstName);
    printf("Enter middle name: ");
    scanf("%49s", middleName);
    printf("Enter last name: ");
    scanf("%49s", lastName);
    combineNames(firstName, middleName, lastName, fullName);
    do
    {
        printf("\n1. Display full name\n");
        printf("2. Find length of full name\n");
        printf("3. Copy full name\n");
        printf("4. Compare first and last name\n");
        printf("5. Display name in reverse order\n");
        printf("6. Count characters\n");
        printf("7. Search a word\n");
        printf("8. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice)
        {
            case 1:
                printf("Full name: %s\n", fullName);
                break;
            case 2:
                findLength(fullName, &length);
                printf("Length of full name: %zu\n", length);
                break;
            case 3:
                copyName(fullName, copiedName);

```

```
    printf("Copied name: %s\n", copiedName);
    break;
case 4:
    compareNames(firstName, lastName, &comparisonResult);
    if (comparisonResult == 0)
        printf("First name and last name are equal.\n");
    else
        printf("First name and last name are different.\n");
    break;
case 5:
    reverseNameOrder(firstName, middleName, lastName, reverseName);
    printf("Reverse name: %s\n", reverseName);
    break;
case 6:
    countCharacters(fullName, &vowels, &consonants, &digits, &spaces);
    printf("Vowels: %d\n", vowels);
    printf("Consonants: %d\n", consonants);
    printf("Digits: %d\n", digits);
    printf("Spaces: %d\n", spaces);
    break;
case 7:
    printf("Enter word to search: ");
    scanf("%49s", word);
    searchWord(fullName, word, &searchResult);
    if (searchResult == 1)
        printf("Word found in full name.\n");
    else
        printf("Word not found in full name.\n");
    break;
case 8:
    printf("Program ended.\n");
    break;

default:
    printf("Invalid choice.\n");
```

```

    }
}
while (choice != 8);
return 0;
}

```

5. Compilation & Execution Evidence

```

kali@kali: ~/Desktop
$ gedit question.c

** (gedit:2053): WARNING **: 13:38:03.021: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2053): WARNING **: 13:38:03.021: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

kali@kali: ~/Desktop
$ gcc question.c -o question

kali@kali: ~/Desktop
$ ./question
Enter first name: Aarav
Enter middle name: Kumar
Enter last name: Sharma

1. Display full name
2. Find length of full name
3. Copy full name
4. Compare first and last name
5. Display name in reverse order
6. Count characters
7. Search a word
8. Exit
Enter your choice: 1
Full name: Aarav Kumar Sharma

1. Display full name
2. Find length of full name
3. Copy full name
4. Compare first and last name
5. Display name in reverse order
6. Count characters
7. Search a word
8. Exit
Enter your choice: 2
Length of full name: 18

1. Display full name
2. Find length of full name
3. Copy full name

```

```

4. Compare first and last name
5. Display name in reverse order
6. Count characters
7. Search a word
8. Exit
Enter your choice: 3
Copied name: Aarav Kumar Sharma

1. Display full name
2. Find length of full name
3. Copy full name
4. Compare first and last name
5. Display name in reverse order
6. Count characters
7. Search a word
8. Exit
Enter your choice: 5
Reverse name: Sharma Kumar Aarav

1. Display full name
2. Find length of full name
3. Copy full name
4. Compare first and last name
5. Display name in reverse order
6. Count characters
7. Search a word
8. Exit
Enter your choice: 6
Vowels: 7
Consonants: 9
Digits: 2
Spaces: 2

1. Display full name
2. Find length of full name
3. Copy full name
4. Compare first and last name
5. Display name in reverse order
6. Count characters
7. Search a word
8. Exit

```

```
Session Actions Edit View Help
Enter your choice: 7
Enter word to search: Kumar
Word found in full name.

1. Display full name
2. Find length of full name
3. Copy full name
4. Compare first and last name
5. Display name in reverse order
6. Count characters
7. Search a word
8. Exit
Enter your choice: 7
Enter word to search: Singh
Word not found in full name.

1. Display full name
2. Find length of full name
3. Copy full name
4. Compare first and last name
5. Display name in reverse order
6. Count characters
7. Search a word
8. Exit
Enter your choice: 8
Program ended.

(kali@kali)-[~/Desktop]
└─$ ./question
Enter first name: Aman
Enter middle name: Raj
Enter last name: Aman

1. Display full name
2. Find length of full name
3. Copy full name
4. Compare first and last name
5. Display name in reverse order
6. Count characters
7. Search a word
8. Exit
```

```
Session Actions Edit View Help
5. Display name in reverse order
6. Count characters
7. Search a word
8. Exit
Enter your choice: 8
Program ended.

(kali@kali)-[~/Desktop]
└─$ ./question
Enter first name: Aman
Enter middle name: Raj
Enter last name: Aman

1. Display full name
2. Find length of full name
3. Copy full name
4. Compare first and last name
5. Display name in reverse order
6. Count characters
7. Search a word
8. Exit
Enter your choice: 6
Vowels: 5
Consonants: 6
Digits: 0
Spaces: 2

1. Display full name
2. Find length of full name
3. Copy full name
4. Compare first and last name
5. Display name in reverse order
6. Count characters
7. Search a word
8. Exit
Enter your choice: 8
Program ended.

(kali@kali)-[~/Desktop]
└─$
```

6. Verification of All Test Cases

Test	Input	Actual/Verified Output	Result
1	Aarav, Kumar, Sharma; choice 1	Aarav Kumar Sharma	Pass
2	Aarav, Kumar, Sharma; choice 2	Length = 18	Pass
3	Aarav, Kumar, Sharma; choice 3	Copied correctly	Pass
4	Aman, Raj, Aman; choice 4	Names equal	Pass

5	Aarav, Kumar, Sharma; choice 5	Sharma Kumar Aarav	Pass
6	Aarav, Kumar, Sharma; choice 6	7 vowels, 9 consonants, 0 digits, 2 spaces	Pass
7	Aarav, Kumar, Sharma; choice 7; Kumar	Word found	Pass
8	Aarav, Kumar, Sharma; choice 7; Singh	Word not found	Pass
9	Aarav, Kumar, Sharma; choice 8	Program ended	Pass

Header Files Used

stdio.h, string.h, ctype.h